Unit 9, Lesson 6: Describing Rigid Transformations

Benchmarks

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

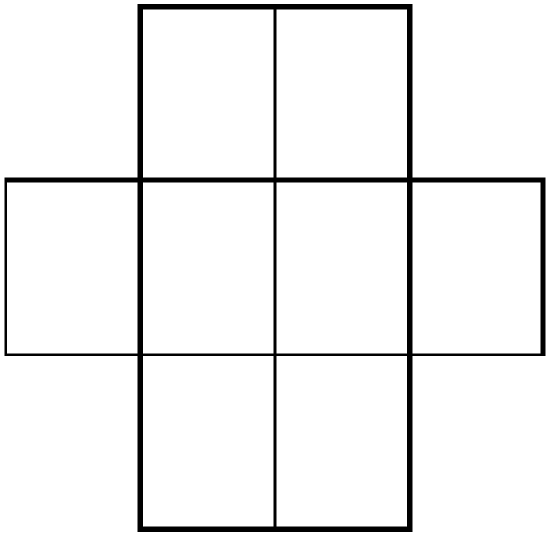
MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

Learning Targets

- ☐ **I can describe the effect of a single transformation.**
- ☐ **I can describe the effect of a transformation on the coordinate plane.**

9.6.1 Warm-Up: Eight-Digit Puzzle

1. Place the numbers 1 through 8 in the rectangles below so that no two consecutive numbers are next to each other horizontally, vertically, or diagonally.



9.6.2 Collaboration: Transformation Matching Cards

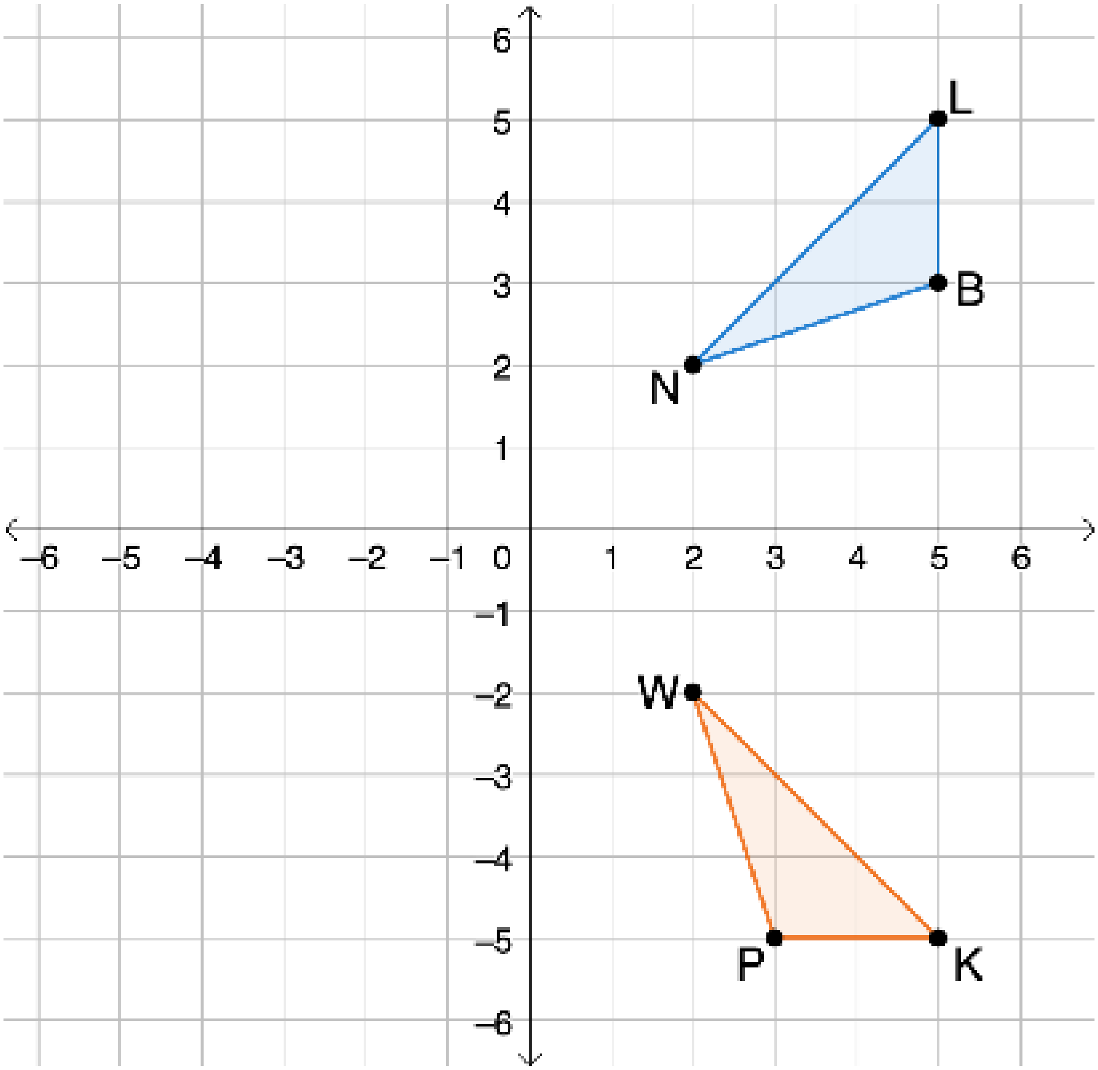
1. **Work with your partner to match the transformation cards with the descriptions of the transformations. Record your matches in the table using the letter shown on the cards.**

Transformation Card	Description Card
1	
2	
3	
4	
5	
6	
7	
8	

2. **Write a brief summary of what you noticed about the location and orientation of the image in comparison to the preimage for each type of transformation.**

Transformation	Location	Orientation
Translation		
Reflection		
Rotation		

3. **Nigel and Elena are discussing what single transformation took place to map ΔBNL to ΔPWK.**

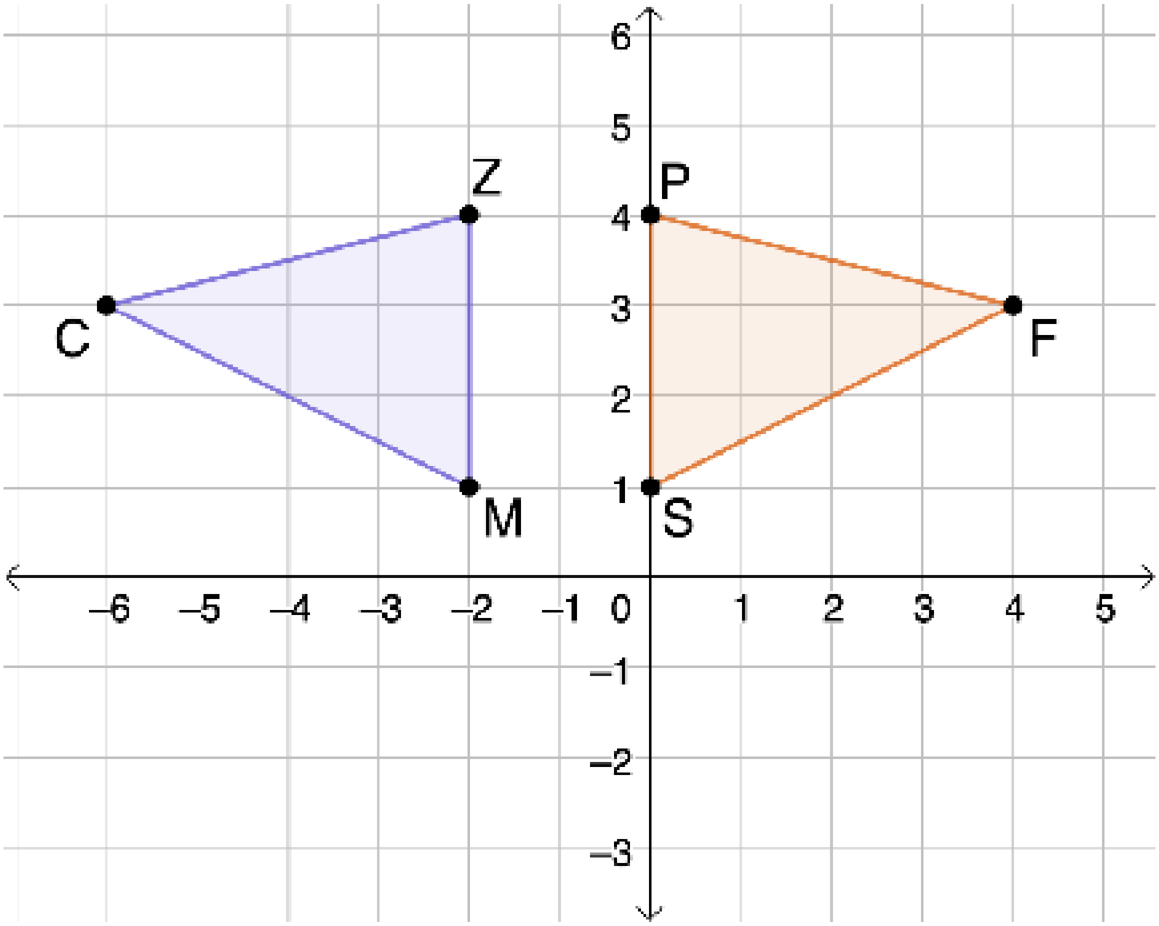


a. **Nigel claims that the transformation is a rotation about the origin. Elena says the transformation is a reflection. Which student do you agree with?**

b. **If you agree with Nigel, what is the angle of rotation and direction? If you agree with Elena, what is the line of reflection?**

9.6.3 Guided Practice: Describing Transformations

1. On the coordinate grid shown, ΔZCM is the preimage, and ΔPFS is the image after a single transformation.



Select the statement that best describes the transformation. Then, complete the chosen statement.

- ☐ ΔZCM can be mapped to ΔPFS by translating it _____ units to the right.
- ☐ ΔZCM can be mapped to ΔPFS by reflecting it over _________________.
- ☐ ΔZCM can be mapped to ΔPFS by rotating it ______° clockwise about the origin.

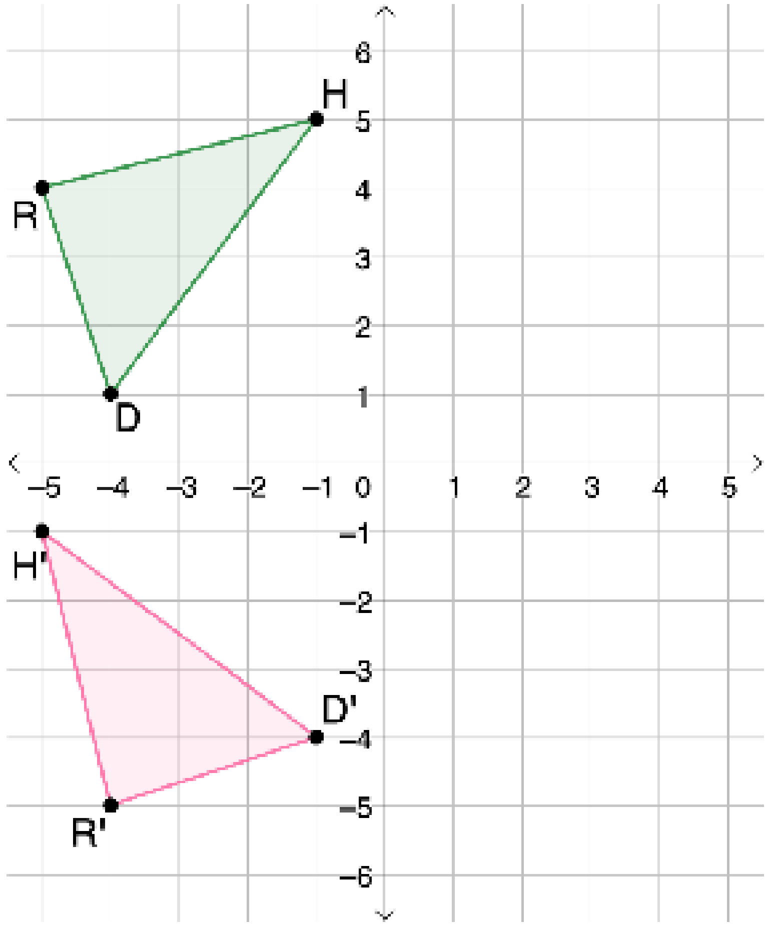
When describing a transformation on the coordinate plane, the description should include the type of transformation as well as the following information.

- **Translation**: the direction and number of units moved
- **Rotation** about the origin: the angle and direction of the rotation

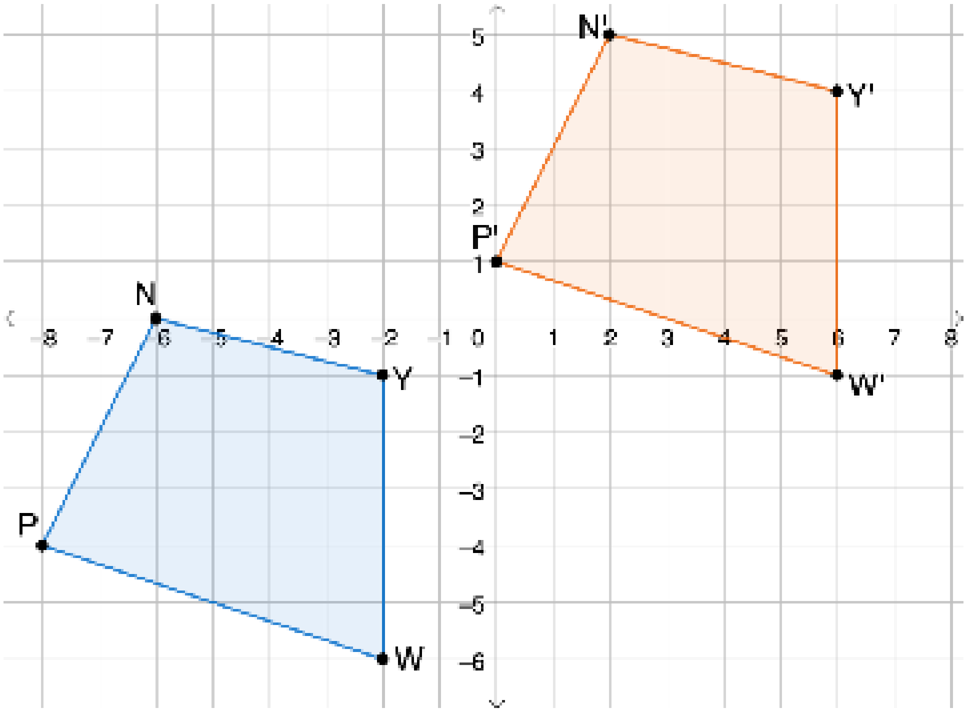
2. **Describe each transformation shown.**

<table>
<tr><th>Transformation</th><th>Description</th></tr>
<tr><td>

ΔHDR is the preimage and $\Delta H'D'R'$ is the image.

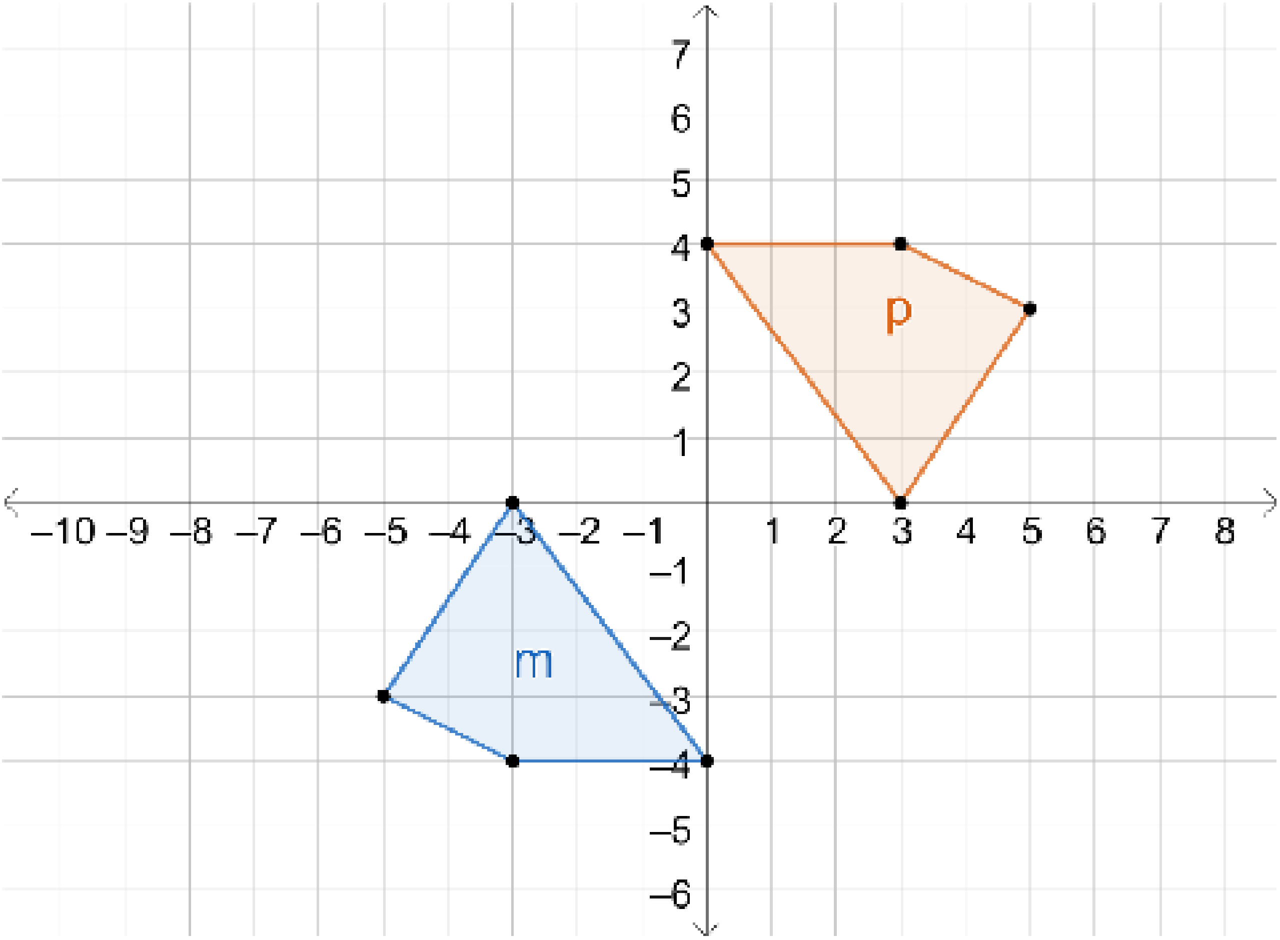


</td><td></td></tr>
</table>

Transformation	Description
Quadrilateral *NYWP* is the preimage and quadrilateral *N'Y'W'P'* is the image. 	

9.6.4 Your Turn!

1. On the coordinate grid, figure *p* is the result of a transformation on figure *m*.



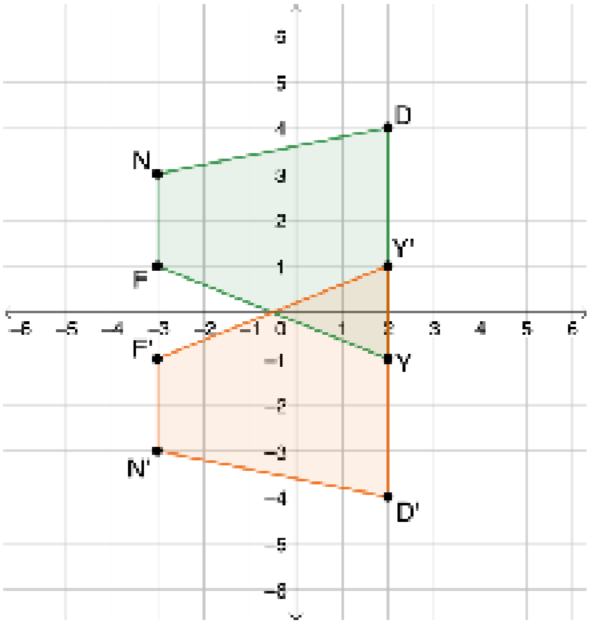
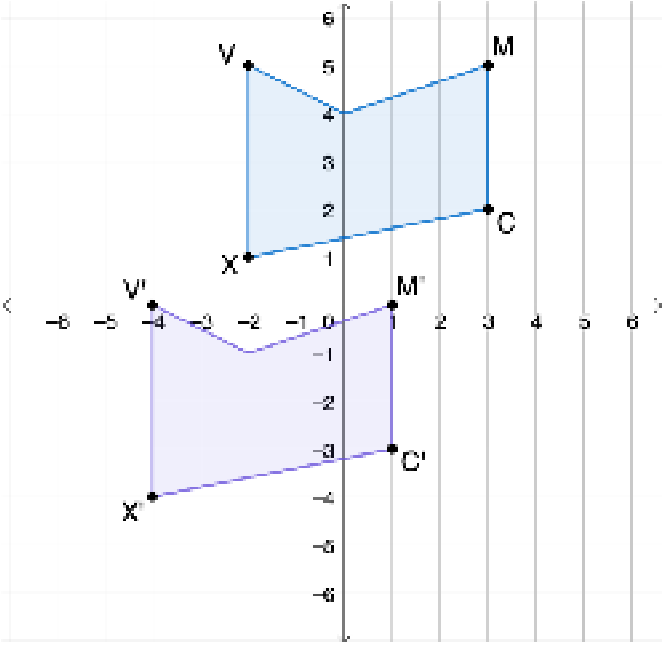
Select the statement that best describes the transformation. Then, complete the chosen statement.

☐ Figure *m* was translated _____ units (○ right ○ left) and _____ units (○ up ○ down) to map to figure *p*.

☐ Figure *m* was reflected to map to figure *p*. The line of reflection is _______________.

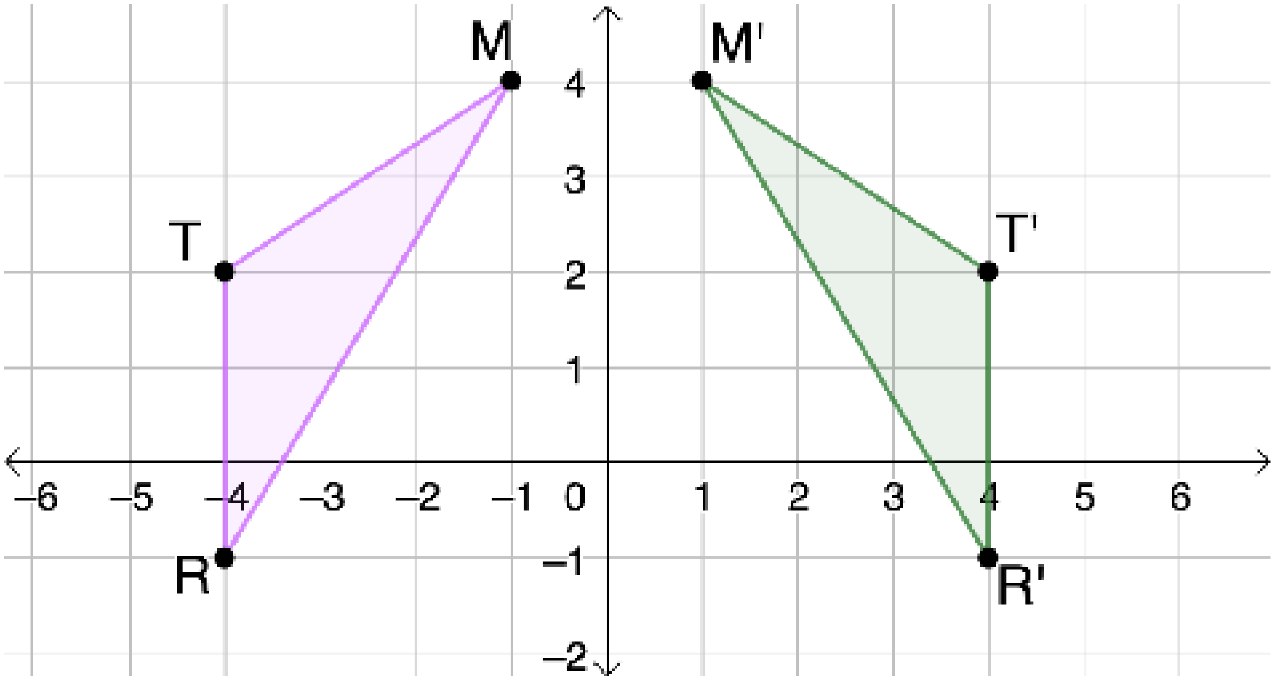
☐ Figure *m* was rotated about the origin to map to figure *p*. The angle of rotation was _____° counterclockwise.

2. **Describe each transformation. Include the units of translation, line of reflection, or angle and direction of rotation as needed.**

Transformation	Description
Quadrilateral $NDYF$ is the preimage and quadrilateral $N'D'Y'F'$ is the image. 	
Quadrilateral $VMCX$ is the preimage and quadrilateral $V'M'C'X'$ is the image. 	

9.6.5 Wrap-Up: Ticket Out the Door

1. ΔMTR, the preimage, and $\Delta M'T'R'$, the image, are shown on the coordinate plane.



Select and complete one statement to describe the transformation.

- ☐ To create the image, the preimage was translated ______ units right.
- ☐ To create the image, the preimage was reflected across the __________________.
- ☐ To create the image, the preimage was rotated ______° clockwise about the origin.